Report on Bookstore Users' Review Data Analysis

Executive Summary

This data analysis project employs pre-processing techniques, machine learning algorithms, and statistical analysis to extract insights from a large dataset gathered from an online bookstore. Data visualization, cleaning, and integration revealed significant patterns and trends, such as the popularity of specific book genres and user age distribution. Machine learning algorithms, linear regression and decision tree regression, were used to predict book evaluations with reasonable accuracy. The findings indicate a predisposition for storytelling and fiction among users aged 20–40. However, there are certain drawbacks, such as missing user data and inherent biases in the analysis. Improvement recommendations include a more in-depth examination of user demographics, book categories, and geographic preferences to improve the recommendation system and maximize commercial strategies. Overall, this research gives practical insights that can help the bookshop improve its performance and better adapt to consumer preferences.

Introduction

As a data scientist, it is crucial to handle data with pre-data processing techniques, Machine Learning and statistical analysis methods. These methods are essential as it directly impact the analysis of a collection of datasets, in order to derive actionable insights that can inform decision-making.

The dataset provided for this data analysis project is a comprehensive collection of information from an online bookstore, including three distinct CSV files: 'BX-Books.csv', 'BX-Users.csv', and 'BX-Ratings.csv'. The 'BX-Books.csv' file contains details on 18,185 books, featuring their unique International Standard Book Number (ISBN), Title, Author, Year of publication, and Publisher. Meanwhile, the 'BX-Users.csv' file offers data on 48,299 users of the online platform, including their unique IDs, City, State, Country, and Age. Lastly, the 'BX-Ratings.csv' file provides insights into user interactions with the books, showcasing user IDs, book ISBNs, and corresponding ratings.

The dataset offers a rich opportunity for exploratory analysis, recommendation system development. The project aims to spot trends and patterns in user preferences. Find out what features influences the popularity of books and how satisfied readers are, to maximize sales, make suitable recommendations to customers for future buying based the findings. As a result, the bookstore's overall performance and profitability can be enhanced.

Specifically, the research question of the project: "**Which book genres tend to receive higher ratings from users among the top 6 most popular states and how do these ratings correlate with other features between the reviewers and books?**" indicates that the analytical results are based on where the users come from, corresponding to the themes of book they read and the age group they belong to.

Methodology

1. Data Visualization.

Data from the input files can be formatted with the imported pandas library. Different types of graphs like bar chart can be plotted by the function in the mathplotlib.pyplot library to visualize data. Thus, the pattern of the distribution of data can be observed and provide useful information for the data analysis. Moreover, some illogical data can be found out.

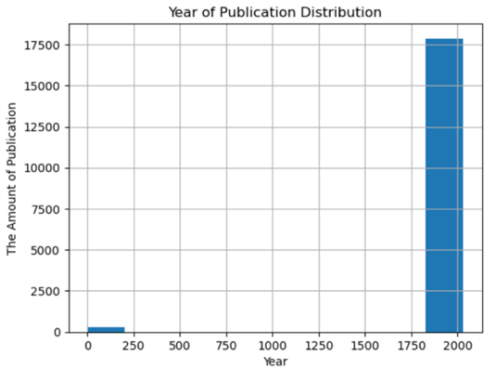


Figure 1: Visualized data from Year-Of-Publication column in BX-Books.csv.

For instance, after visualizing the dataset of year of publication (Figure 1), an error is found – abnormal years recorded between integral 0-250, which are apparently the outliers that should be removed.

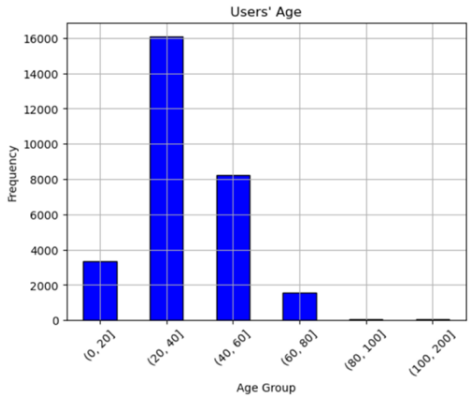


Figure 2: Visualized data from users-country column in BX-Users.csv.

One more example from another input file, the age of the users is visualized by plotting a histogram through mathplotlib.pyplot (Figure 2), after dropping the empty items and grouping them into serval age groups with the support from the pandas library. It is easy to tell that most users who give feedback after purchasing are in the age group (20, 40]. It is also unreasonable to have ages between (100, 200].

2. Data Pre-processing.

Data cleaning is a foundational element of data science, as it helps ensure the accuracy and reliability of the data being analysed. This process involves identifying and correcting errors, inconsistencies, and missing values in the data. Additionally, techniques such as integration, reduction, transformation, normalization, de-noising, and dimensionality reduction are also included (Lötsch et al., 2022).

2.1 Data imputation.

It is spotted that there are many empty cells in the csv files, some users did not fill in their country, many of them left their age blank.

```
User-ID            0
User-City         61
User-State         0
User-Country     669
User-Age       18952
dtype: int64
```

Figure 3: The amount of missing data in BX-Users.csv.

After counting the exact number of missing values in the input file via function in pandas library (Figure 3), some changes should be made or else will affect the accuracy of the analysis of users' country, which is the main attribute of the project, while the missing city data will not have significant impact in the analysis due to the small amount and it is not the main direction to be explored in this project. In view of this, the empty country cells are replaced by checking the state of the same users, filling in the corresponding country the state belongs to using previous record, and simply ignore the missing city data.

Despite there are quite a lot of missing data in users' age, as the project focuses more on the location of users, the lost numbers are simply replaced by -100 (Figure 4) to indicate that it is not suitable to include the data while analysing.

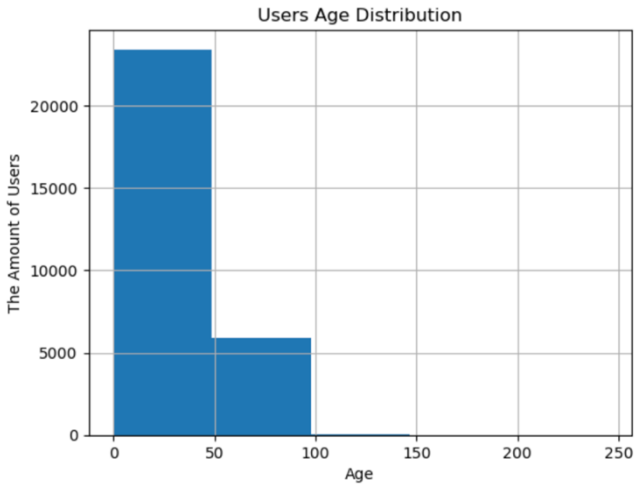


Figure 4: Age of users.

2.2 Fixing Wrong Data.

"Wrong data" does not have to mean "empty cells" or "wrong format", it might simply be incorrect, such as if someone entered "200" instead of "20" for age. Looking at the data set can sometimes reveal incorrect facts since there are expected ranges.

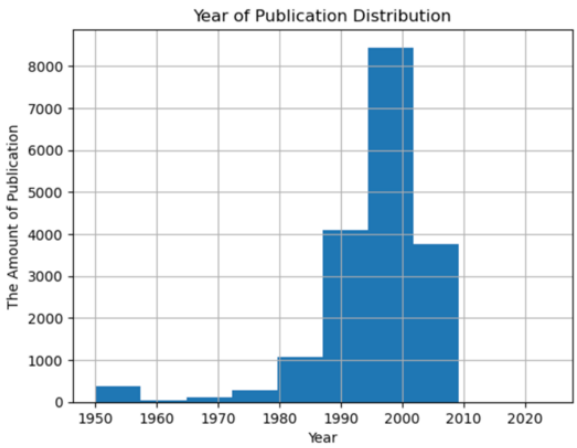


Figure 5: Modified year of publication dataset.

In the data visualization part, error is found in the year of publication dataset (Figure 1). To fix the problem, limit is set, the year of publication should be between 1950 and 2024 (Figure 5).

2.3 Scaling.

Scaling the data to a specific range can ensure fair comparisons.

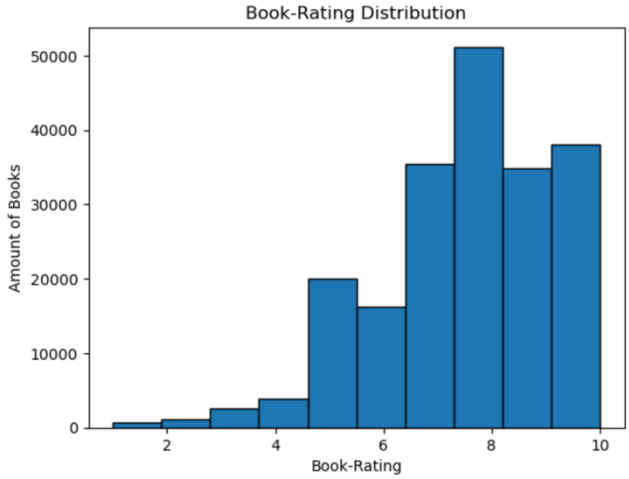
1. 

Figure 6: Visualized book rating dataset.

It is evidence from the bar graph (Figure 6) that the dataset has a negative skew, illustrating that in general, users tend to give ratings between 7-9. To scale the ratings for better analysis, they are divided to 5 groups (1, 2, 3, 4, 5). 1 represent ratings between 1-6, while 2, 3, 4, 5 represent ratings 7, 8, 9 ,10 respectively.

Furthermore, the record of age is divided into groups 1, 2, 3, 4, 5 representing the age group (0, 20], (20, 40], (40, 60], (60, 80], (80, 100] respectively, while the of year of publication for the books is grouped by [1950,1960), [1960,1970), [1970,1980), [1980,1990), [1990,2000), [2000,2010), [2010,2020), [2020,2030).

2.4 Data Reshaping.

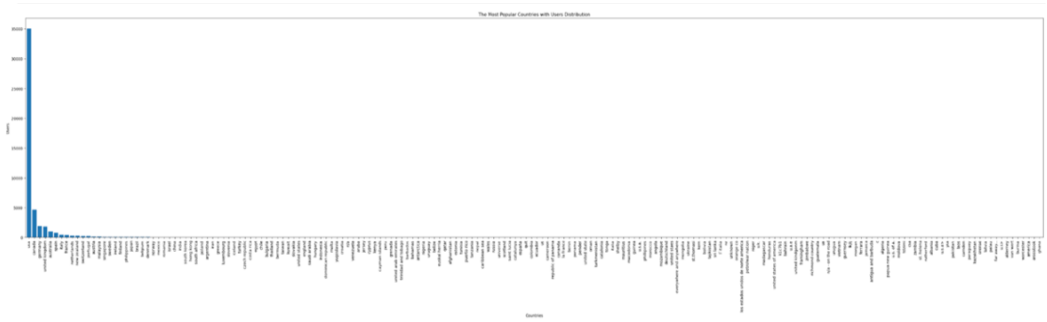


Figure 7: Data visualized from users-country column in BX-Users.csv before reshaping.

From the graph (Figure 7), it is apparent that the distribution of countries is uneven and there are many countries with low frequency, some even just have one to two users. Compared to other countries like the US, Canada, Germany, these countries have no influence towards the topic of the project. In view of this, countries with users fewer than 800 will be removed as these countries with low frequency do not cause much impact to the statistical result.

After manipulation of the country dataset, the 5 most popular countries can be found, which made it one step closer to the main part of the project – finding the 6 most popular states of user that gave high ratings and analysis the relation between the location and high ratings.

2.5 Dimensionality reduction.

Reducing the number of variables or features in the data can simplify complex data and improve computational efficiency, such as using techniques like principal component analysis or feature selection, enables researchers to focus on the most important variables and remove any irrelevant or redundant information.

	ISBN	Year-Of-Publication	Year-Of-Publication-Bin	Book-Author-Encoded	Book-Publisher-Encoded	embedding
0	0002005018	2001	5	5190	522	-0.479524
1	0374157065	1999	4	2169	406	-0.346551
2	0399135782	1991	4	240	952	0.010049
3	0440234743	1999	4	3079	287	-0.488134
4	0452264464	1994	4	6132	910	-0.334188

Figure 8: Final form of book.

The method can be illustrated by the example (Figure 8), three kinds of information are compressed to one column. The column embedding included the books' title, author and publisher.

Moreover, in the merged output from section 2.6(Figure 9), when the original attributes from the BX-Books.csv file included the book title, author, year of publication, publisher and the BX-Users.csv file included the users' ID, city, stage, country, age. The new table drops the users' ID and city, and also the ISBN of books.

2.6 Data Integration.

Integration is the step of combining data from different sources or formats to create a unified dataset for analysis, a method for achieving a comprehensive and accurate understanding of complex research problems. By integrating different types of data, researchers can gain deeper insights into the research question at hand.

	Book-Rating	Year-Of-Publication	Year-Of-Publication-Bin	Book-Author-Encoded	Book-Publisher-Encoded	embedding	User-Age	User-Age-Bin	User-State-Encoded	User-Country-Encoded
0	7	2001	5	2556	325	-0.483261	-100	-100	133	178
1	10	2001	5	2556	325	-0.483261	-100	-100	489	101
2	3	1999	4	5762	1062	-0.238919	-100	-100	489	101
3	1	2002	5	3459	1018	-0.389705	-100	-100	489	101
4	9	2003	5	1841	1222	-0.356511	-100	-100	489	101
...	...	...	...	...	...	...	...	...	...	...
204159	10	1998	4	1512	456	-0.382565	-100	-100	859	57
204160	9	2003	5	1856	54	-0.290446	25	2	859	57
204161	10	2003	5	1856	54	-0.290446	-100	-100	833	147
204162	6	1998	4	1112	850	0.125248	24	2	133	178
204163	9	1998	4	1112	850	0.125248	-100	-100	859	178

204164 rows × 10 columns

Figure 9: Table showing data from book dataset merging user and rating dataset.

Merging the dataset from different files by their foreign key (ISBN of the books), resulting in a single table (Figure 9) for better and easier data exploration. Meanwhile, some attributes are not included in the new table (section 2.5).

2.7 Data Type Conversion.

	User-ID	User-Age	User-Age-Bin	User-State-Encoded	User-Country-Encoded
0	8	-100	-100	715	27
1	9	-100	-100	942	178
2	16	-100	-100	639	178
3	17	-100	-100	1012	178
4	19	14	1	0	178

Figure 10: User Table with new modified attributes.

This involves changing non-numerical data to numerical which allows easier and more efficient dataset comparison (Figure 10).

3. Machine Learning.

In this project, two models are implemented – Linear Regression and Decision Tree Regression. Below is a brief introduction of the models.

3.1 Linear Regression.

Linear regression is a machine learning algorithm that is widely used for predictive analysis. It is a powerful tool for modelling the relationship between a dependent variable and one or more independent variables and predicting the results (Anushree et al., 2019).

3.2 Decision Tree Regression.

Decision tree regression is another common machine learning algorithm used for predicting continuous variables. It works by partitioning the feature space into several simple regions, and then fitting a simple model, like a constant, in each region (Anushree et al., 2019).

3.3 RMSE & KNN classification.

The RMSE score is commonly used as an evaluation metric for regression models, measuring the average difference between predicted and actual values. On the other hand, KNN classification is a method that falls under the umbrella of instance-based learning. It aims to classify new data points based on their proximity to the nearest neighbours in the training data. Both the RMSE score and KNN classification are valuable tools in the field of machine learning (Plevris et al., 2022).

Data Exploration and Analysis

After all the data pre-processing procedures, the datasets can now be used for exploration.

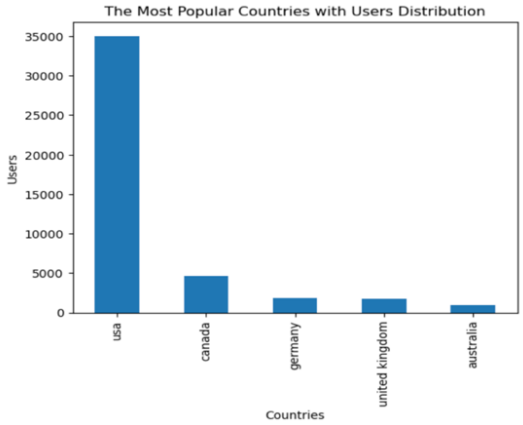


Figure 11: Top 5 countries.

As mentioned above, after reducing the massive dataset (Figure 11) of users' country by keeping countries with higher than 800 users, the top 5 countries are as follow: USA, Canada, Germany, UK, Australia.

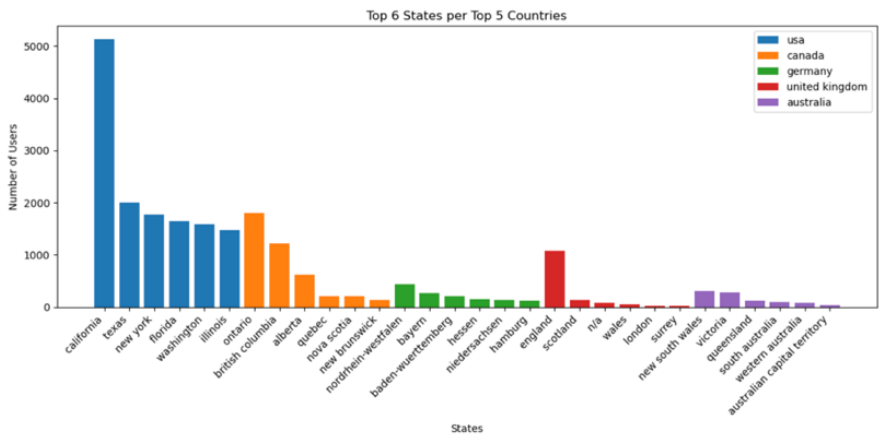


Figure 12: Top 6 states in the top 5 countries where the users give higher ratings.

Country	1st Popular State	2nd Popular State	3rd Popular State	4th Popular State	5th Popular State	6th Popular State
USA	California	Texas	New York	Florida	Washington	Illinois
Canada	Ontario	British Columbia	Alberta	Quebec	Nova Scotia	New Brunswick
Germany	Nordrhein	Bayern	Baden-wuerttemberg	Hessen	Niedersachsen	Hamburg
UK	England	Scotland	N/A	Wales	London	Surrey
Australia	NSW	Victoria	Queensland	South Australia	Western Australia	Australian Capital Territory

Table 1: Top 6 states in the top 5 countries where the users give higher ratings.

Among the countries with higher number of users, their top 6 states with the most users are generated (Figure 12) and recorded (Table 1). Noted that the remain states with lower frequency are grouped by new category "other states" to enhance time complexity.

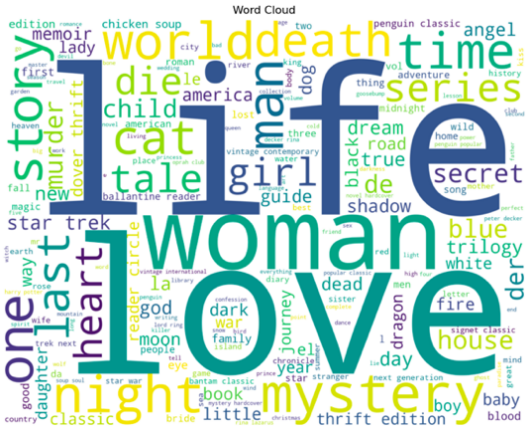


Figure 13: Word cloud of book titles.

The word cloud (Figure 13) reflects the common topic among the popular books. Obviously, among the users from the top 6 states, books theme regarding life, love is more popular and more likely to receive good reviews.

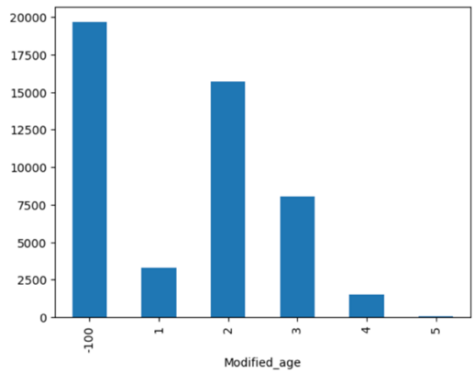


Figure 14: Processed age of users.

Most of the users are aged between 20 to 40, marked as 2 (Figure 14).

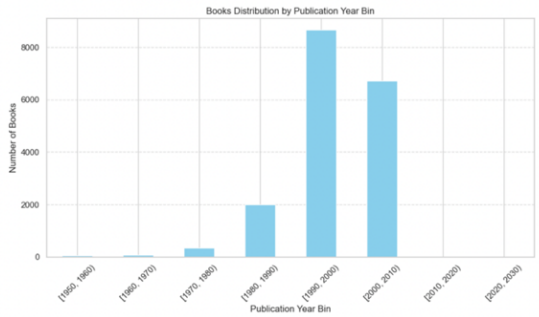


Figure 15: Distribution of Publication of books binned.

The bar graph (Figure 15) shows that the books being sold are mainly after 1990s.

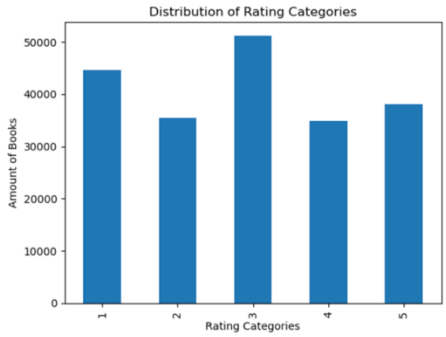


Figure 16: Distribution of ratings vs amount of books.

The ratings are now grouped by 5 categories after scaling (Figure 16). Users tend to give ratings higher than 6 for most books, with 8 having the highest frequency.

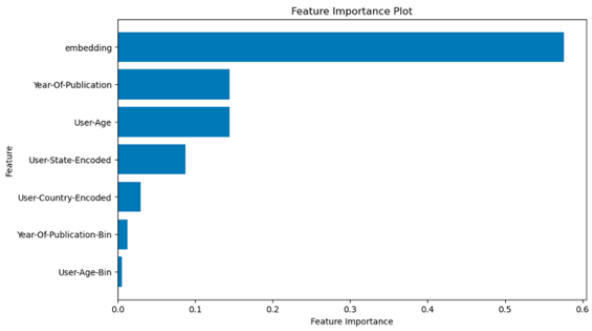


Figure 17: Features of affecting ratings.

The depicted figure (Figure 17) indicates that the main feature affecting the ratings of the book are the theme, followed by the year of publication and the users' age.

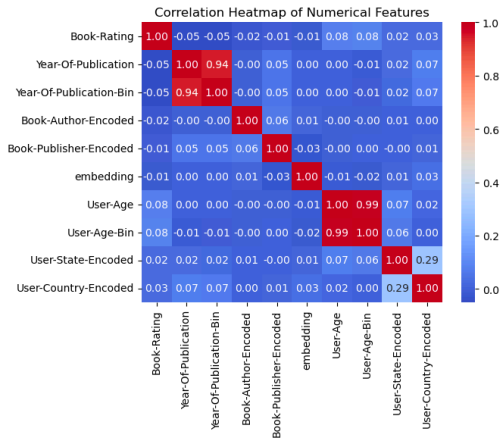


Figure 18: Correlation heatmap of numerical features.

According to the heat map (Figure 18), we can see the correlation among some factors while some of them are raw some of them are modified, while red colour indicates highly related relationship. It is told that the factors, "year-of-publication", "book- author-encoded", "embedding", "user-age", "user-state-encoded" and "user-country-encoded", are low independent. Hence, they would not affect each other. In addition, the raw and the processed dataset years of books publication and age of users, are high related, which means that when we process the data, these major attributes should have fewer changes.

```
Linear Regression Cross-Validation RMSE Scores:
[1.7531925  1.81144026 1.75442448 1.80234744 1.79961167]
Linear Regression Mean RMSE: 1.7842032717220033
Linear Regression Cross-Validation R^2 Scores:
[ 0.0025787   0.01211977  0.00920443  0.00229984 -0.00807367]
Linear Regression Mean R^2: 0.003625815348235517
```

Figure 19: Linear regression: RMSE and R^2 scores.

The linear regression results (Figure 19), show that the root mean square error (RMSE) and r squared (R^2). Root mean squared error is defined as the average difference between actual values and model predictions. The results show that there is a variance of roughly 1.784 ratings. As a result, when the model predicts user ratings, it may estimate within a deviation. For example, if a user gives an actual rating of 7, the model will estimate between 5.216 and 8.784. R squared is a measurement that provides information on the goodness of a model. Furthermore, applying linear regression with 0.004 r squared indicates that the linear regression line does not properly fit the actual ratings.

```
Decision Tree Regression Cross-Validation RMSE Scores:
[2.45761625 2.48098995 2.42463929 2.41622535 2.46301057]
Decision Tree Regression Mean RMSE: 2.4484962811644677
Decision Tree Regression Cross-Validation R^2 Scores:
[-0.95996163 -0.85313166 -0.89238271 -0.79307264 -0.88828359]
Decision Tree Regression Mean R^2: -0.8773664444958664
```

Figure 20: Decision tree regression: RMSE and R^2 scores.

The decision tree regression model uses a tree-like description of decisions and their potential consequences. From the graph (Figure 20), the RMSE score indicates that there is a deviation of 2.448 when estimating user ratings. In addition, we can see that the r square is -0.877, indicating that the data in the model may not have a connection. When comparing these two models, it can be concluded that linear regression is more suitable to reveal the association between components and user ratings using RMSE and r squared. In terms of RMSE, the linear regression model has lower deviation for ratings. For one rating, there is a 0.664 difference between two models.

```
KNN Classifier Metrics:
Accuracy: 0.2206
F1-score: 0.2076
Classification Report:
              precision    recall  f1-score   support

           1       0.26      0.29      0.28      8785
           2       0.18      0.18      0.18      7173
           3       0.26      0.26      0.26     10360
           4       0.19      0.18      0.18      6930
           5       0.25      0.23      0.24      7585

    accuracy                           0.23     40833
   macro avg       0.23      0.23      0.23     40833
weighted avg       0.23      0.23      0.23     40833
```

Figure 21: KNN classification: accuracy and F1-score.

In terms of obtaining sufficient decision tree regression results, a model with classification is employed. A classification model tries to predict the correct label for a given input data. KNN classification predicts the grouping of individual data points based on proximity. According to the graph (Figure 21), it would cluster the factors that may affect users' ratings and determine their means. However, the model's accuracy is low, with just approximately 22.1% of correct predictions. Besides, when dealing with vast volumes of unbalanced data, the F1-score combines accuracy and recall, indicating that it is inaccurate at 0.2076.

Results

After extensive data pre-processing, the dataset was explored to identify trends and patterns. The data analysis provided valuable insights into user demographics, book preferences, and factors influencing ratings.

To sum up, the top 5 countries with the highest number of users were identified: USA, Canada, Germany, UK, and Australia, while the top 6 states within these countries where users gave higher ratings were also listed. A world cloud analysis revealed common themes among popular books, with topics revolving around life and love being prominent. Majority of users fell within the age range of 20 to 40, indicating a younger demographic. Analysis of book publication years showed that most books being sold were from the 1990s onwards. Users tend to give ratings higher than 6, with 8 being the most frequent rating. Factors affecting book ratings were identified, with theme being the primary influencer followed by year of publication and user age. A heat map illustrated correlations among various factors, highlighting relationships between attributes like year of publication and user age.

Meanwhile, Linear regression is proved to be a more effective model for predicting ratings compared to decision tree regression and KNN classification.

It is accurate enough to predict the review of a new book. Whenever a new book is available for selling, implementing the model to predict the book may have a higher popularity among which states of different countries. This allows the bookstore to decide whether the book is worth recommending and where to promote it more.

Discussion and Interpretation

- Lack of Asian users.
  The number of users who is in the US is much higher than other countries. This may because the bookstore is based in the US, thus most users of the system is from the US, Canada and other countries in Europe. Living in the Globalisation era, it is suggested to also put some effort in promoting the store in Asia, results in a more professional and worldwide business.
- Novel, fictions are more common.
  Love, life, mystery, story are the most common theme of books according to the world cloud generated in the previous section. It is reasonable that people enjoy reading storytelling and non-informative books more. So, focusing more on these books will be a wise move. At the same time, collect the search history of the users and promote the non-fiction books related to their needs can boost the sale of unpopular books.
- Target users' age group.

Focus more on users aged between 20-40 as they are the largest group of customers. As per the big influence of the age and year of publication towards the customers' review, for the recommendation system, it is suggested that recommending books in a specific published year to readers in a specific age group.

- Small amount of new book.
  It is spotted that most of the books are from 1990-2000. Adding more books published recently can also be considered and this allows users to gain up-to-date perspective of the world through reading newer books.
- Users' age is an important feature.
  Knowing the users' age can result in more related recommendation of books. In this project, many ages from the users are missing. The system should make sure that users are filling in their age. However, it may cause privacy issue. Therefore, instead of setting their birth date as a compulsory information, birthday discount can be offered to encourage users providing their ages.
- High average rating.
  This indicates that in general, the store is selling books that satisfy users. To maintain its quality, such choices of books should be referred to whenever new books are ready to be imported.
- Standard for highly recommended books.
  As mentioned above, most of the books received good comments. For the recommendation system, it is suggested that the new books should be highly recommended if the predicted rating is 9/10.

Limitations and improvement opportunities

For data constraints, other key features such as users' gender, and occupation which may also affect their choices towards the themes of books are not included in the provided dataset. Also, there are too many data lost for age of user. There may be some potential biases during the analysis. Specifically, the relation between the ratings and book features are all obtained from the reviews from users who are in the top 6 states. It is assumed that users in the same states will have the same preferences. For future improvement, it is suggested to go deeper so that the city of the users can also be analysed. Apart from the features of the users, the authors of the books can also be explored. Moreover, the books can be grouped into different categories and find out the more popular books in each category, corresponding to the users who buy the same kind of books often and recommend the books in terms of the book genres. What's more, we can divide users with ages, and we can focus on age with different interests for book genres, and then the recommendations system would recommend specific books to such users. Finally, the raw data and the data after the pre-processing process are hoped to have a smaller difference.

Conclusion

In short, this data analysis project is a comprehensive effort to extract actionable insights from a large dataset collected by an online bookstore. Using a methodical approach that included data visualization, cleansing, integration, and machine learning approaches, we discovered significant patterns and trends that might drive strategic decision-making for the bookstore's operations.

The pre-processing phase highlighted important characteristics of the dataset, including inconsistencies in data inputs such as incorrect publication years and missing age data. The integrity and reliability of subsequent analyses is assured by managing missing data carefully and detecting outliers. Furthermore, the integration of diverse data sources enabled a comprehensive picture of relation between users and books.

Machine learning techniques, notably linear regression and decision tree regression are being utilized as effective tools for predicting book ratings and finding significant factors influencing user satisfaction. While linear regression performed better in terms of accuracy and interpretability, decision tree

regression revealed extra information about the complexity of user preferences, both models emphasized the importance of different factors.

Although the absence of certain variables like users' gender, restricts the depth of the understanding of user preferences. Additionally, some potential bias may oversimplify user behaviours. Looking ahead, there are various option to refine and expand our analysis. Future research could go deeper into users and books categorization, enabling more personalized suggestions. Furthermore, including new data sources, may improve our understanding of user-book interactions and preferences.

References

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